

Chirp Mass Scaling from the Quadrupole Formula

A Step-by-Step Derivation for a Gravitational-Wave School

1 Assumptions and Setup

We derive the leading-order (Newtonian orbit + lowest-order quadrupole radiation, i.e. “0PN”) relation between the gravitational-wave frequency evolution $\dot{f}$ of a compact binary and its *chirp mass* $\mathcal{M}$. This is the relation LIGO/Virgo/KAGRA use to read off $\mathcal{M}$ directly from the observed inspiral waveform.

Stated assumptions:

1. Two point masses m_1, m_2 on a **circular** orbit ($e = 0$). Radiation reaction circularizes eccentric binaries well before merger (Peters 1964), so this is the generic late-inspiral limit.
2. **Newtonian orbital dynamics:** weak field and slow motion for the *orbit itself*, $v/c \ll 1$, valid away from merger.
3. The system radiates according to the **quadrupole formula** (leading term of the multipole expansion of linearized GR in TT gauge, valid for a slow-moving, weak-field source of size $\ll \lambda_{GW}$), taken here as the known input result:

$$P_{GW} \equiv \left. \frac{dE}{dt} \right|_{\text{rad}} = \frac{G}{5c^5} \left\langle \ddot{Q}_{ij} \ddot{Q}^{ij} \right\rangle, \quad (1)$$

where $Q_{ij} = \int \rho(\mathbf{x}) \left(x_i x_j - \frac{1}{3} \delta_{ij} r^2 \right) d^3x$ is the trace-free (reduced) mass quadrupole moment.

4. **Adiabatic approximation:** the radiation-reaction timescale is much longer than the orbital period, so at each instant the system is well described as an instantaneous circular orbit of slowly shrinking separation $a(t)$.
5. Define total mass $M = m_1 + m_2$, reduced mass $\mu = m_1 m_2 / M$, symmetric mass ratio $\eta = \mu / M$, and **chirp mass**

$$\mathcal{M} \equiv \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} = \mu^{3/5} M^{2/5}. \quad (2)$$

The two-body problem in the center-of-mass frame reduces to a single fictitious particle of mass μ orbiting at the relative separation $a = |\mathbf{x}_1 - \mathbf{x}_2|$.

2 Quadrupole Moment of a Circular Binary

Place the orbital plane in the x - y plane. With orbital angular frequency $\omega \equiv \omega_{\text{orb}}$,

$$x(t) = a \cos(\omega t), \quad y(t) = a \sin(\omega t), \quad z(t) = 0, \quad r^2 = x^2 + y^2 + z^2 = a^2 \text{ (const)}. \quad (3)$$

The reduced quadrupole moment of the effective one-body system (mass μ at $\mathbf{x}(t)$) is

$$Q_{ij}(t) = \mu \left(x_i x_j - \frac{1}{3} \delta_{ij} a^2 \right). \quad (4)$$

Using $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$ and $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$:

$$Q_{xx} = \mu a^2 \left(\cos^2 \omega t - \frac{1}{3} \right) = \mu a^2 \left[\frac{1}{6} + \frac{1}{2} \cos 2\omega t \right], \quad (5)$$

$$Q_{yy} = \mu a^2 \left(\sin^2 \omega t - \frac{1}{3} \right) = \mu a^2 \left[\frac{1}{6} - \frac{1}{2} \cos 2\omega t \right], \quad (6)$$

$$Q_{zz} = \mu a^2 \left(0 - \frac{1}{3} \right) = -\frac{1}{3} \mu a^2 \quad (\text{constant}), \quad (7)$$

$$Q_{xy} = Q_{yx} = \mu a^2 \cos \omega t \sin \omega t = \frac{1}{2} \mu a^2 \sin 2\omega t, \quad (8)$$

$$Q_{xz} = Q_{zx} = Q_{yz} = Q_{zy} = 0. \quad (9)$$

One checks $\text{Tr } Q_{ij} = \frac{1}{6} + \frac{1}{6} - \frac{1}{3} = 0$ as required for the trace-free moment. Note the orbital motion sources radiation at *twice* the orbital frequency — this is the origin of the well-known $f_{GW} = 2f_{\text{orb}}$ relation for the dominant quadrupole ($\ell = 2, m = 2$) mode.

3 Third Time Derivative

Only the oscillatory ($2\omega t$ -dependent) pieces survive differentiation; constants drop out. Using $\frac{d^3}{dt^3} \cos 2\omega t = 8\omega^3 \sin 2\omega t$ and $\frac{d^3}{dt^3} \sin 2\omega t = -8\omega^3 \cos 2\omega t$:

$$\ddot{Q}_{xx} = 4\mu a^2 \omega^3 \sin 2\omega t, \quad (10)$$

$$\ddot{Q}_{yy} = -4\mu a^2 \omega^3 \sin 2\omega t, \quad (11)$$

$$\ddot{Q}_{zz} = 0, \quad (12)$$

$$\ddot{Q}_{xy} = \ddot{Q}_{yx} = -4\mu a^2 \omega^3 \cos 2\omega t. \quad (13)$$

4 Radiated Power

With flat spatial indices, $\ddot{Q}_{ij} \ddot{Q}^{ij} = \sum_{i,j} (\ddot{Q}_{ij})^2$:

$$\begin{aligned} \ddot{Q}_{ij} \ddot{Q}^{ij} &= (\ddot{Q}_{xx})^2 + (\ddot{Q}_{yy})^2 + (\ddot{Q}_{zz})^2 + 2(\ddot{Q}_{xy})^2 \\ &= 16\mu^2 a^4 \omega^6 \sin^2 2\omega t + 16\mu^2 a^4 \omega^6 \sin^2 2\omega t + 2(16\mu^2 a^4 \omega^6 \cos^2 2\omega t) \\ &= 32\mu^2 a^4 \omega^6 (\sin^2 2\omega t + \cos^2 2\omega t) = 32\mu^2 a^4 \omega^6. \end{aligned} \quad (14)$$

This is exactly time-independent — no averaging is even required, a nice feature of the circular orbit: the luminosity is constant in time (all the “chirping” comes from the adiabatic drift of a and ω , not short-timescale oscillation of P). Plugging into Eq. (1):

$$P_{GW} = \frac{32}{5} \frac{G}{c^5} \mu^2 a^4 \omega^4 \omega^2 = \frac{32}{5} \frac{G}{c^5} \mu^2 a^4 \omega^6. \quad (15)$$

5 Eliminating a : Kepler's Third Law

For Newtonian gravity, the circular-orbit relation between separation and angular frequency is

$$\omega^2 = \frac{GM}{a^3} \implies a = \left(\frac{GM}{\omega^2} \right)^{1/3}. \quad (16)$$

Then $a^4 = (GM)^{4/3} \omega^{-8/3}$, and Eq. (15) becomes

$$P_{GW} = \frac{32}{5} \frac{G}{c^5} \mu^2 (GM)^{4/3} \omega^{-8/3} \omega^6 = \frac{32}{5} \frac{G^{7/3}}{c^5} \mu^2 M^{4/3} \omega^{10/3}. \quad (17)$$

Now use the chirp mass identity implied by Eq. (2): since $\mathcal{M} = \mu^{3/5} M^{2/5}$,

$$\mathcal{M}^{10/3} = \left(\mu^{3/5} M^{2/5} \right)^{10/3} = \mu^2 M^{4/3}, \quad (18)$$

so that

$$\boxed{P_{GW} = \frac{32}{5} \frac{G^{7/3}}{c^5} \mathcal{M}^{10/3} \omega^{10/3}}. \quad (19)$$

This is the key result: *the radiated power depends on the two masses only through the single combination $\mathcal{M}$ (at this order).*

6 Orbital Energy and Energy Balance

The Newtonian orbital energy of a circular binary (kinetic + potential, virial theorem $T = -\frac{1}{2}U$) is

$$E_{\text{orb}} = -\frac{GM\mu}{2a}. \quad (20)$$

Substituting Eq. (16), $a = (GM/\omega^2)^{1/3}$:

$$E_{\text{orb}}(\omega) = -\frac{GM\mu}{2} \left(\frac{\omega^2}{GM} \right)^{1/3} = -\frac{\mu}{2} (GM)^{2/3} \omega^{2/3}. \quad (21)$$

Energy balance (adiabatic radiation reaction): the orbit loses energy at exactly the rate it is radiated away,

$$\frac{dE_{\text{orb}}}{dt} = -P_{GW}. \quad (22)$$

Differentiating Eq. (21) via the chain rule,

$$\frac{dE_{\text{orb}}}{dt} = \frac{dE_{\text{orb}}}{d\omega} \frac{d\omega}{dt} = -\frac{\mu}{3} (GM)^{2/3} \omega^{-1/3} \dot{\omega}. \quad (23)$$

Setting this equal to $-P_{GW}$ from Eq. (19) and solving for $\dot{\omega}$:

$$\frac{\mu}{3} (GM)^{2/3} \omega^{-1/3} \dot{\omega} = \frac{32}{5} \frac{G^{7/3}}{c^5} \mathcal{M}^{10/3} \omega^{10/3} \quad (24)$$

$$\dot{\omega} = \frac{96}{5} \frac{G^{7/3}}{c^5} \frac{\mathcal{M}^{10/3}}{\mu (GM)^{2/3}} \omega^{11/3}. \quad (25)$$

Simplify the mass prefactor using $\mathcal{M}^{10/3} = \mu^2 M^{4/3}$ (Eq. 18):

$$\frac{\mathcal{M}^{10/3}}{\mu(GM)^{2/3}} = \frac{\mu^2 M^{4/3}}{\mu G^{2/3} M^{2/3}} = \mu M^{2/3} G^{-2/3}, \quad (26)$$

and note the second chirp-mass identity, $\mathcal{M}^{5/3} = (\mu^{3/5} M^{2/5})^{5/3} = \mu M^{2/3}$, so

$$\frac{\mathcal{M}^{10/3}}{\mu(GM)^{2/3}} = \mathcal{M}^{5/3} G^{-2/3}. \quad (27)$$

Therefore

$$\dot{\omega}_{\text{orb}} = \frac{96}{5} \frac{G^{5/3}}{c^5} \mathcal{M}^{5/3} \omega_{\text{orb}}^{11/3}. \quad (28)$$

7 The Chirp Equation in Terms of GW Frequency

The dominant ($\ell = 2, m = 2$) gravitational-wave frequency is twice the orbital frequency (as seen already in Sec. 2: Q_{ij} oscillates at 2ω), $f_{GW} = 2f_{\text{orb}} = \omega_{\text{orb}}/\pi$, i.e. $\omega_{\text{orb}} = \pi f_{GW}$. Substituting into Eq. (28) (and using $\dot{\omega}_{\text{orb}} = \pi \dot{f}_{GW}$):

$$\pi \dot{f}_{GW} = \frac{96}{5} \frac{G^{5/3}}{c^5} \mathcal{M}^{5/3} (\pi f_{GW})^{11/3} \quad (29)$$

$$\dot{f}_{GW} = \frac{96}{5} \pi^{8/3} \left(\frac{GM}{c^3} \right)^{5/3} f_{GW}^{11/3}. \quad (30)$$

This is *the* chirp-mass scaling law: the rate of change of the GW frequency depends on the two component masses *only through* $\mathcal{M}$, and scales as $f^{11/3}$ — hence the audible “chirp” rising in both pitch and rate as $f \rightarrow \infty$ (formally) at coalescence.

Inverting for the chirp mass (data-analysis application)

Since f_{GW} and $\dot{f}_{GW}$ are both directly measurable from the observed strain (e.g., by tracking the instantaneous frequency of the time-domain signal or the stationary-phase template), Eq. (30) can be inverted:

$$\mathcal{M} = \frac{c^3}{G} \left[\frac{5}{96} \pi^{-8/3} f_{GW}^{-11/3} \dot{f}_{GW} \right]^{3/5}. \quad (31)$$

This is precisely how the chirp mass of GW150914 and subsequent events was first measured: purely from the shape of the frequency evolution, with *no* independent knowledge of m_1 and m_2 separately.

8 Time-to-Coalescence and Separation Scaling

(a) Frequency vs. time to merger. Equation (30) is separable: $f^{-11/3} df = \frac{96}{5} \pi^{8/3} (GM/c^3)^{5/3} dt$. Integrating from time t (frequency f) to the formal coalescence time t_c (where $f \rightarrow \infty$),

$$\int_f^\infty f'^{-11/3} df' = -\frac{3}{8} f^{-8/3} = \frac{96}{5} \pi^{8/3} \left(\frac{GM}{c^3} \right)^{5/3} (t_c - t) \Big|_{\text{sign fixed by } f > 0}, \quad (32)$$

giving the standard **time-to-coalescence** formula

$$\tau(f) \equiv t_c - t = \frac{5}{256} \pi^{-8/3} \left(\frac{G\mathcal{M}}{c^3} \right)^{-5/3} f^{-8/3}. \quad (33)$$

This lets you estimate, e.g., how long a binary spends above the 10 Hz band of a ground-based detector, purely from $\mathcal{M}$.

(b) Separation vs. time (Peters 1964). It is also instructive to derive the analogous ODE directly for $a(t)$ rather than $\omega(t)$. From $E_{\text{orb}} = -GM\mu/(2a)$, $dE_{\text{orb}}/da = GM\mu/(2a^2)$. Using Kepler's law to write $\omega^6 = (GM)^3/a^9$ in Eq. (15),

$$P_{GW} = \frac{32}{5} \frac{G}{c^5} \mu^2 a^4 \frac{(GM)^3}{a^9} = \frac{32}{5} \frac{G^4}{c^5} \frac{\mu^2 M^3}{a^5}. \quad (34)$$

Energy balance $\left(\frac{dE_{\text{orb}}}{da} \right) \dot{a} = -P_{GW}$ gives

$$\frac{GM\mu}{2a^2} \dot{a} = -\frac{32}{5} \frac{G^4}{c^5} \frac{\mu^2 M^3}{a^5} \implies \boxed{\dot{a} = -\frac{64}{5} \frac{G^3 m_1 m_2 M}{c^5 a^3}}. \quad (35)$$

This is the celebrated Peters (1964) inspiral-rate formula for a circular orbit. Separating and integrating from $a_0 \equiv a(0)$,

$$a(t)^4 = a_0^4 - \frac{256}{5} \frac{G^3 \mu M^2}{c^5} t, \quad (36)$$

so the **merger time** (formally where $a \rightarrow 0$) is

$$\boxed{t_{\text{merge}} = \frac{5}{256} \frac{c^5 a_0^4}{G^3 \mu M^2}}. \quad (37)$$

Note the strong scaling $t_{\text{merge}} \propto a_0^4/M^3$ (roughly, for comparable masses): halving the initial separation shrinks the merger time by a factor of 16 — this is why GW-driven mergers of stellar-mass compact objects can occur within a Hubble time only once a_0 is astrophysically small (comparable to $\sim$ solar-radius scales or less), while wide binaries essentially never merge from GW emission alone.

9 Summary of Results

$$P_{GW} = \frac{32}{5} \frac{G^{7/3}}{c^5} \mathcal{M}^{10/3} \omega_{\text{orb}}^{10/3}, \quad \dot{f}_{GW} = \frac{96}{5} \pi^{8/3} \left(\frac{G\mathcal{M}}{c^3} \right)^{5/3} f_{GW}^{11/3}, \quad (38)$$

$$\dot{a} = -\frac{64}{5} \frac{G^3 m_1 m_2 M}{c^5 a^3}, \quad \tau(f) = \frac{5}{256} \pi^{-8/3} \left(\frac{G\mathcal{M}}{c^3} \right)^{-5/3} f^{-8/3}, \quad (39)$$

$$\mathcal{M} = \mu^{3/5} M^{2/5} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}, \quad \mathcal{M} = \frac{c^3}{G} \left[\frac{5}{96} \pi^{-8/3} f^{-11/3} \dot{f} \right]^{3/5}. \quad (40)$$